DESIGN OF 1X2 COMPACT UWB MIMO ANTENNA UISNG STAIR CASE & COMBLINE
DIAGRAMS

1. Introduction:

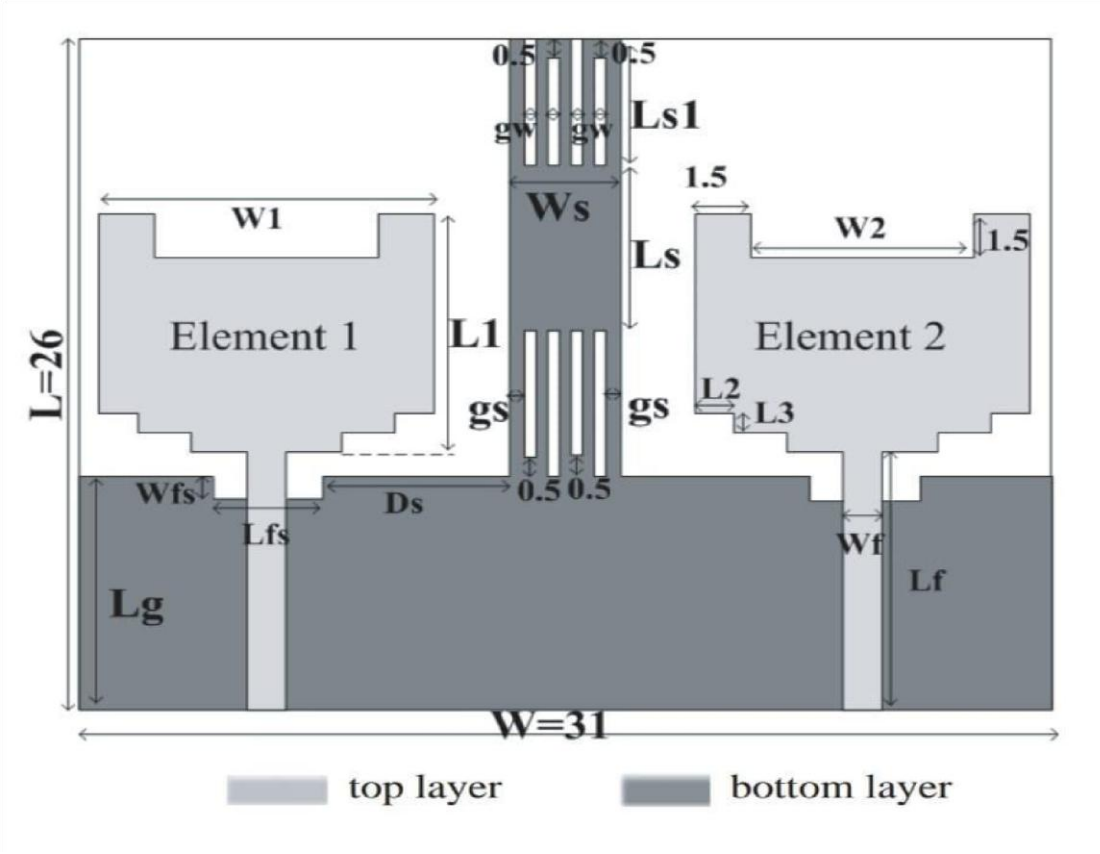


Fig. 1. Geometry of the designed antenna

L	Lfs	Lf	L_1	L_2	L_3	Lg	Ls	Ds	gs
26	4	9	8	1.5	0.5	8	7	6	0.45
W	Wfs	Wf	W_1	W_2	W_3	W_4	Ws	Ls_1	gw
31	0.8	1.4	11	8	5	8	3	5	0.3

Dimensions of the designed antenna (mm)

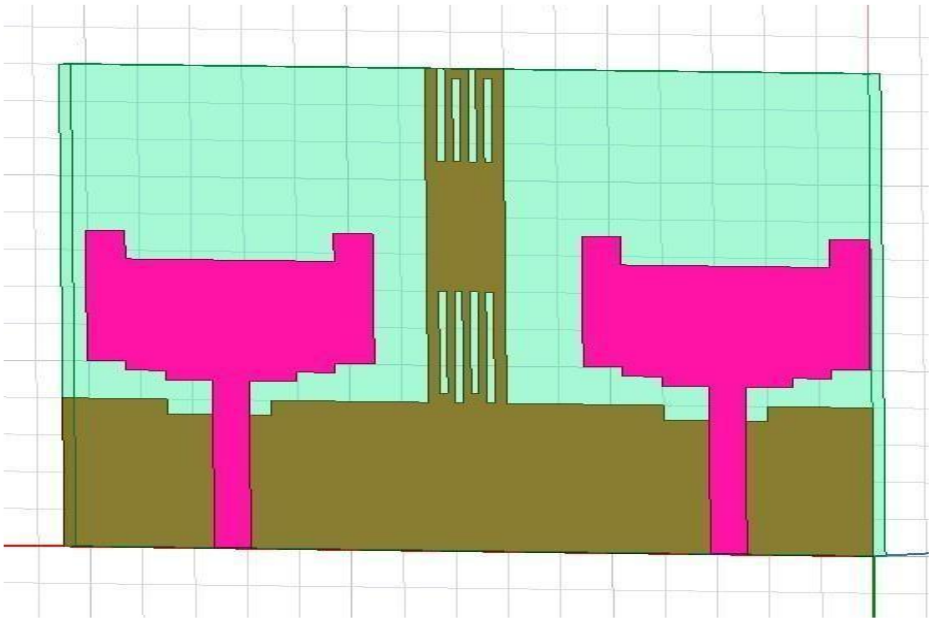


Fig 2: UWB MIMO Antenna Design

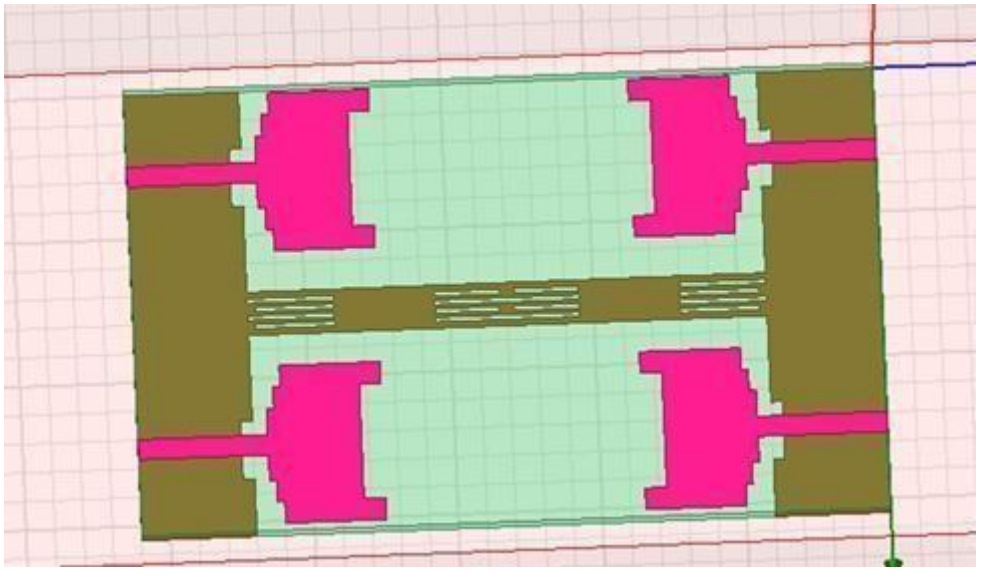


Fig 2.1: 1x2 UWB MIMO Antenna Design

4.Reflection Coefficient (S11) Analysis :

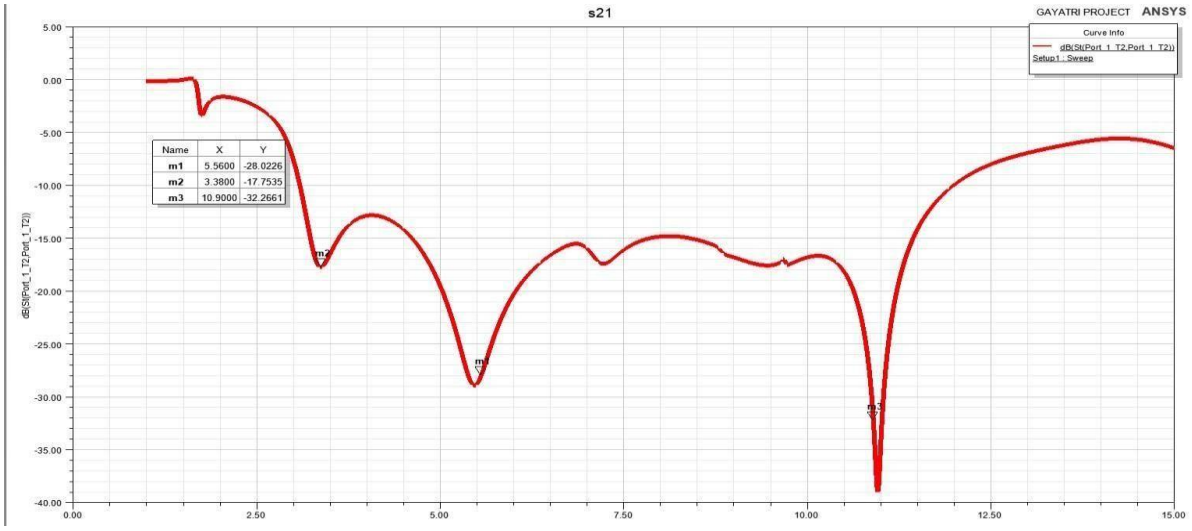


Fig 3 : S11 Parameter for first patch

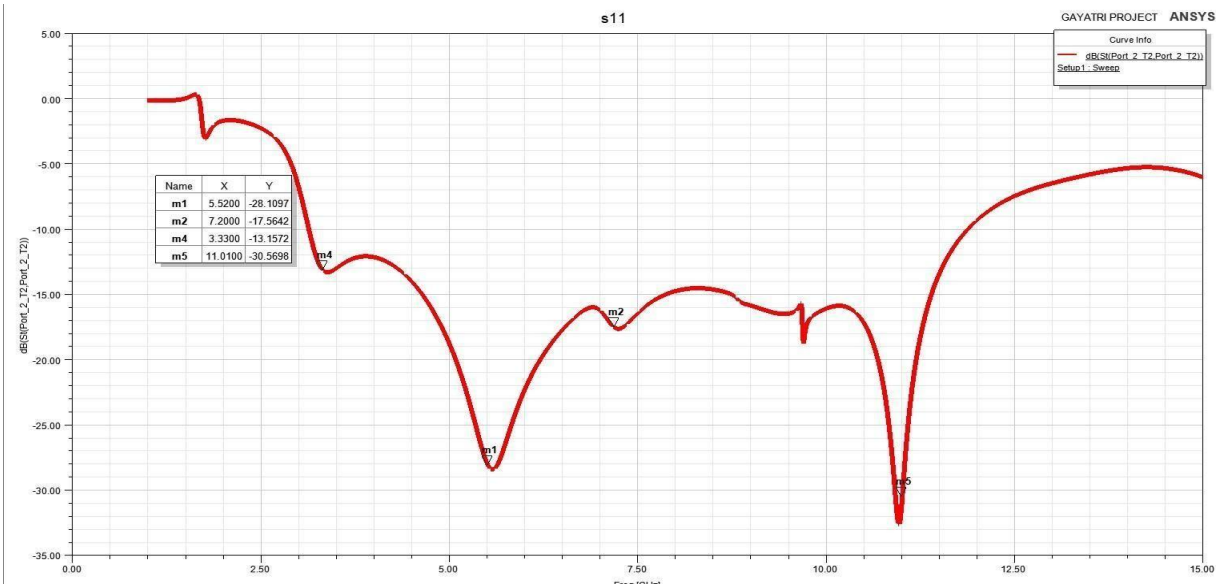


Fig 4 : S22 Parameter for second patch

Isolation (S21) and Mutual Coupling Analysis :

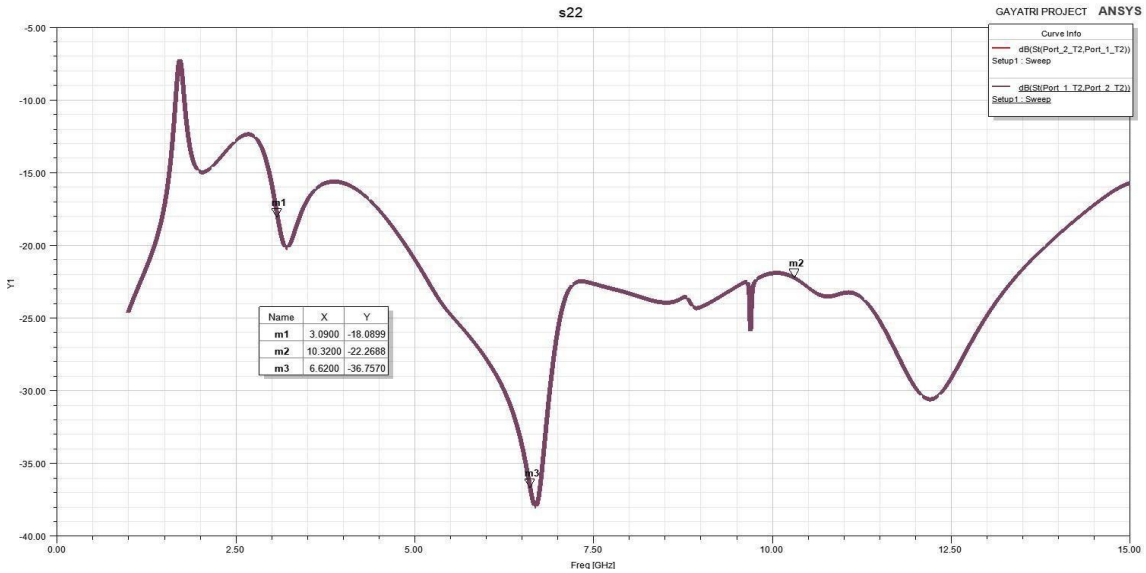


Fig 5: Isolation (S21) and Mutual Coupling Analysis

Radiation Pattern and Far-Field Characteristics :

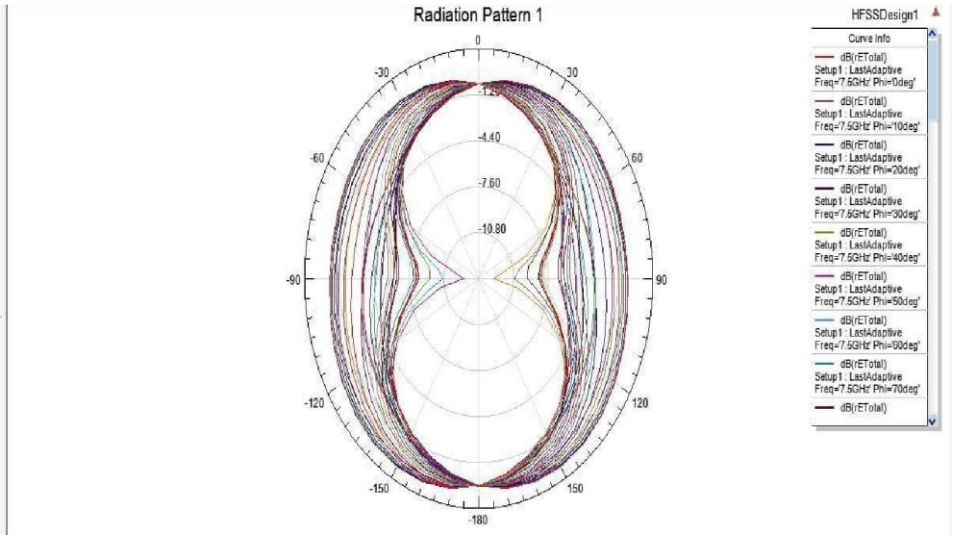


Fig 6: Radiation pattern

Envelope Correlation Coefficient (ECC) and Diversity Performance :

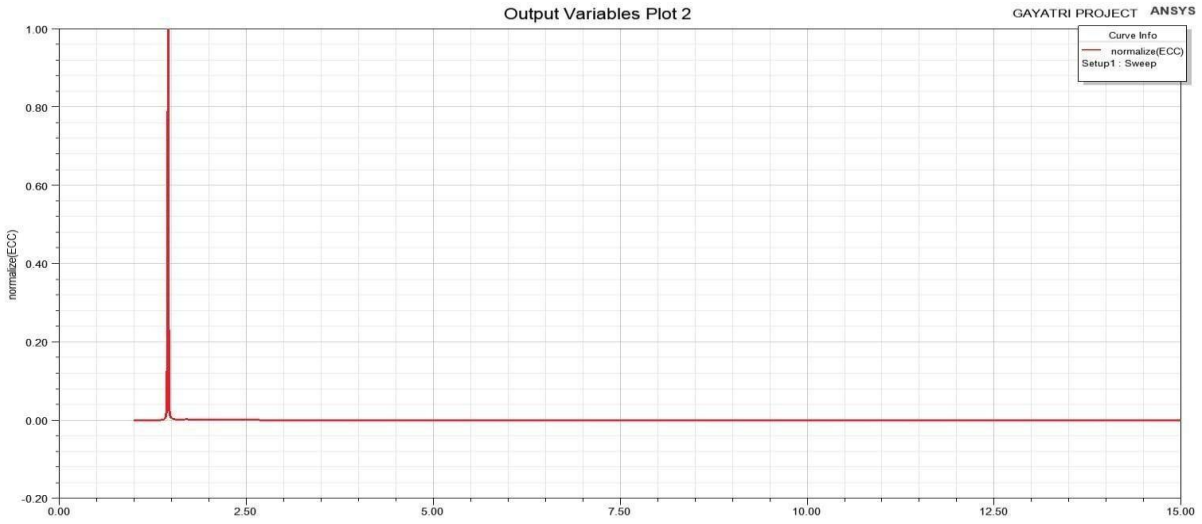


Fig 6: Envelope Correlation Coefficient (ECC)

Channel Capacity Loss :

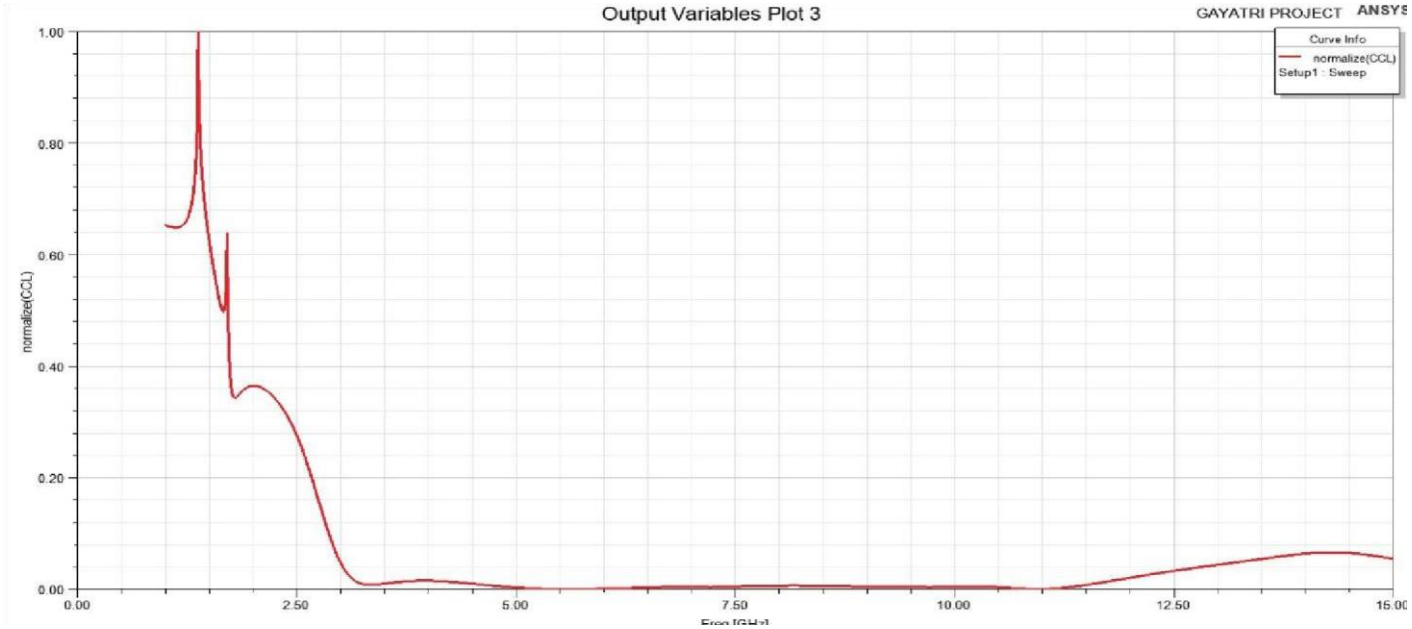


Fig 7:Channel Capacity Loss

Diversity Gain :

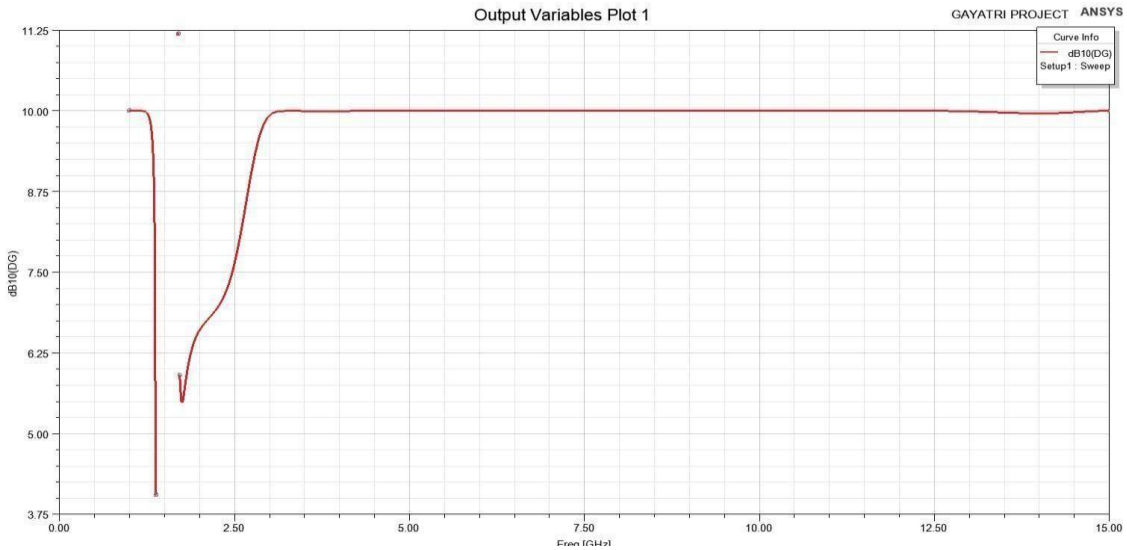


Fig 8: Diversity Gain Conclusion

conclusion

Parameter	Achieved	Expected
Substrate Material	Rogers RT/duroid	Low-loss substrate required
Operating Bandwidth (S11)	3.1–10.6 GHz	Must cover UWB band
Return Loss (S11)	< -10 dB	< -10 dB
Isolation (S21)	< -15 to -20 dB	> -15 dB
ECC	< 0.01	< 0.1
Gain	2-6 dBi	Moderate expected
Efficiency	> 80%	> 70%
Radiation Pattern	Stable omni	Stable required
Diversity Gain	~10 dB	~10 dB ideal
Channel Capacity Loss	< 0.4 bits/s/Hz	< 0.4 bits/s/Hz